

2023

PHYSICS — HONOURS — PRACTICAL

Paper : SEC-1P

Full Marks : 100

The figures in the margin indicate full marks.

SET - 2

[LNB - 20, VIVA - 20, PROGRAMMING - 60]

Answer *any one* question from *each group*.

Group - A

1. Save the following data table in a file : Fit the data points of the second row against that of the

x	1.22	1.44	1.67	1.89	2.11	2.33	2.56	2.78	3.
$f(x)$	5.013	3.65	2.78	2.19	1.77	1.467	1.234	1.054	0.910

first row using GNUPLOT with the equation $f(x) = ax^b$. Write the values of the fitted parameters a , b . Show the fitted curve along with the data points in the same plot. Set the title "Data fitting" and 'x' and 'f(x)' for x-axis and y-axis respectively.

20

2. Plot the following functions for $-1 \leq x \leq 3$ on the same graph using GNUPLOT. Set the range of y-values to be from -2 to 4. Set 1000 samples. Mark each curve with different line style and line widths. Label x-axis and y-axis as 'x-values' and 'y-values' respectively. Set the title of the plot 'Function plots'. Save the plot in a suitable format.

$$f_1(x) = \text{sech}(|x|), \quad f_2(x) = |x|^3 \ln(x)/(4 + \sqrt{x}), \quad f_3(x) = 0.83x^3 - 4.2x^{1/4}$$

20

3. Show the polar plot of the function $-\frac{3}{2}(5\cos^2 t - 1)\sin t$, where t is the angle made with the reference directions (positive x-axis) using GNUPLOT. Set samples to 800. Set polar grid with appropriate angle between gridlines. Take $0 \leq t \leq 2\pi$ and $0 \leq r \leq 1$. Set title of the plot as "Third order associated Legendre polynomial". Save the plot in a suitable format.

20

4. Three functions are defined in the given range as the following :

$$f(x) = 0.71x^2 - 6.29\sqrt{x}, \quad x \geq 4$$

$$g(x) = \sin^2(x^3), \quad 2 \leq x < 4$$

$$h(x) = e^{|x|}, \quad -2 \leq x < 2.$$

Set the range of x values from -2 to $+5$ and the range of y values from -2 to $+10$. Set ticks of the x -axis at an interval of 0.5 and ticks of y -axis at an interval 1.0 . Set title "Piece-wise continuous functions". Plot the three functions using GNUPLOT and store the plot in a suitable format. 20

5. Plot the following polar functions in GNUPLOT :

$$r(\theta) = 2 \cos(2\theta), \quad r(\theta) = \frac{\theta}{3}.$$

Take $0 \leq \theta \leq 2\pi$. Set legend to each function as "Rose petal" and "Spiral of Archimedes" respectively. Set title "Some polar functions". Use different line styles and line widths for the two curves. Save the plot in a suitable format. 20

6. Make the following data table in a file :

x	0.0	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0
y	0.0	0.366	0.424	0.425	0.402	0.368	0.33	0.292	0.255	0.222	0.191

Write a gnuplot script to fit the data with the curve $f(x) = \sqrt{x}e^{-ax}$ with parameter a . Write the value of a . Plot the above data table along with $f(x)$. Set the title "Data fitting" and ' x ' and ' $f(x)$ ' for x -label and y -label respectively. Save the plot in a suitable format. 20

7. Plot the following parametric functions for $0 \leq \theta \leq 2\pi$:

$$f_x(\theta) = (1 + \cos\theta)\cos\theta, \quad f_y(\theta) = (1 + \cos\theta)\sin\theta.$$

Set the title of the plot as "Parametric plot". Set sample size 1000. Save the plot in a suitable format. 20

Group - B

8. (a) Write a python program to find the solution of transcendental equation $x \ln(x) = x$ using bisection method. Find the solution lying in the range $[1, 4]$ with three decimal places of accuracy.
- (b) Write a python program to calculate the sum of the following A.P. series :
- $$S = 1 + 5 + 9 + 13 + \dots \text{ up to 8 terms.}$$
- Print the output.

30+10

9. (a) Write a python program to compute the root of the equation $e^x - \sin(x) = 0$ using Newton-Raphson method with initial value $x = -2$, correct up to three decimal places. Print the root.
- (b) Write a python program to calculate the trace of the following matrix :

$$M = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ -7 & -8 & -9 \end{pmatrix}$$

Print the output.

30+10

10. (a) A first order differential equation is given as below :

$$y \frac{dy}{dx} = x \text{ with } y(1) = 2.$$

Create list (say X) with 1000 x -values between $[1, 2]$. Solve this differential equation using Euler method for x -values in the list X . It is given that the exact solution is

$$y^2 = x^2 + 3$$

Calculate and print the value of y at $x = 2$. Compare your answer with the value from the exact solution.

- (b) Write a python program to calculate the sum of the following series :

$$S = \frac{1.2}{3.4} + \frac{5.6}{7.8} + \dots \text{ up to 4 terms.}$$

Print the output.

30+10

11. (a) Using Newton-Raphson method with starting point $x_0 = 5$, determine the root of the following equation correct up to three decimal places.

$$x^3 - 7x^2 + 8x - 3 = 0.$$

Print the root.

- (b) Write a python program to calculate the transpose of the following matrix :

$$M = \begin{pmatrix} 3 & 4 & -2 \\ 1 & -5 & -1 \\ 2 & 6 & 9 \end{pmatrix}.$$

Print the output.

30+10

12. (a) Acceleration of a particle can be written as

$$v \frac{dv}{dx} = \frac{\sin(x)}{x^2 + 11x + 28}$$

Write a python program to solve the above differential equation using Euler method and find the velocity at $x = 10\text{m}$, provided the initial velocity $v(x = 0) = 1$. Print the output. Take the increment of x as $dx = 0.001$.

- (b) Write a python program to find the sum of the following A.P. series :

$S = 7 + 11 + 15 + \dots$ up to 6 terms.

Print the output.

30+10

13. (a) Put the following numbers in a list :

2.83, - 8.31, - 3.67, 7.46, - 0.73, 1.39, - 7.38, 5.92.

Write a python program to sort these numbers in ascending order. Store and print the sorted list. Create another list with the squares of the previous (sorted) list. Print this new list. Determine the sum of the squared elements of the list and print it.

- (b) Make the following data table in a file. Write a python program to read the columns of this file with two lists say X and Y

-1.0	2.718
-0.8	1.424
-0.6	0.656
-0.4	0.239
-0.2	0.049
0.0	0.0
0.2	0.033

Print the two lists X and Y .

30+10

14. (a) Sort the following integers in ascending order and put them in a list.

11, 28, 17, 5, 19, 15, 7, 31, 23, 13. Print the sorted list.

- (b) Determine the prime numbers from the sorted list and print those numbers.

10+30